Purely Analytic Positivity of the Upward-Closed Scalar Gap

Global middle-ratio analytic module

Abstract

The angular-block reduction for the middle-ratio spherical-shell proxy leaves one smooth scalar inequality. This note proves that inequality on the upward-closed region $\rho_c \le \rho \le 39/50,\ K \ge k_{11}(\rho)$ without interval arithmetic or a finite numerical certificate. The loss integral is first differentiated with respect to the spectral cutoff. A turning-point lower bound and an exact beta integral show that the scalar gap is strictly increasing in that cutoff for every $K > 12$. At the base curve, a rational secant majorant gives a uniform analytic estimate for the loss. The remaining one-variable inequality is then reduced to positivity of a degree-nine polynomial; all of its Bernstein coefficients on one rational interval are displayed and are positive. Thus there is no upper-frequency cutoff in the scalar-positivity theorem.

1 The scalar gap

For $0 \le s \le 1$, put

$$G_1(s) := \frac{1}{\pi}\left(\sqrt{1-s^2} - s\arccos s\right). \tag{1}$$

Fix

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}, \tag{2}$$

and consider

$$\mathcal{R}_c := \left\{(\rho, K) : \rho_c \le \rho \le \frac{39}{50}, \quad K \ge k_{11}(\rho)\right\}. \tag{3}$$

The scalar gap left by the angular-block estimate is

$$\begin{aligned}\mathcal{S}(\rho, K) := \frac{(1+2\rho^2)K^2}{12} - \mathcal{J}(\rho, K) - \frac{\pi\rho K}{4(1-\rho)} + \frac{1}{2}\\ + \frac{2\pi\rho K}{\arccos\rho}\left(\frac{KG_1(\rho)}{4(1+2KG_1(\rho))} + \frac{3}{32}\right),\end{aligned} \tag{4}$$

where

$$\mathcal{J}(\rho, K) := \frac{K^2}{2}\int_\rho^1 \frac{s}{(1+2KG_1(s))^2}\, ds. \tag{5}$$

Theorem 1 (Upward-closed scalar positivity)**.** *For every $\rho_c \le \rho \le 39/50$ and every $K \ge k_{11}(\rho)$,*

$$\boxed{\mathcal{S}(\rho, K) > 0.} \tag{6}$$

We use only the elementary rational estimates

$$\frac{157}{50} < \pi < \frac{22}{7}, \qquad \frac{140}{99} < \sqrt{2}, \qquad \frac{5}{3} < \sqrt{3}. \tag{7}$$

The two square-root estimates follow immediately by squaring; the displayed bounds for π are the classical Archimedean bounds (with the lower one weakened to 157/50).

2 Strict monotonicity in the spectral cutoff

The identity

$$G_1(s) = \frac{1}{\pi}\int_s^1 \arccos u \, du \tag{8}$$

and $1 - \cos v \le v^2/2$ imply

$$G_1(s) \ge \frac{2\sqrt{2}}{3\pi}(1-s)^{3/2}. \tag{9}$$

Indeed, $\arccos u \ge \sqrt{2(1-u)}$, and integration gives (9).

Lemma 2 (Derivative of the boundary-layer loss)**.** *For every fixed* $0 < \rho < 1$ *and every* $K > 0$*,*

$$\partial_K \mathcal{J}(\rho, K) = K \int_\rho^1 \frac{s}{(1+2KG_1(s))^3}\, ds. \tag{10}$$

Moreover, if $K > 12$*, then*

$$\int_\rho^1 \frac{s}{(1+2KG_1(s))^3}\, ds < \frac{88}{567}. \tag{11}$$

Proof. Differentiating (5) under the integral sign gives

$$\begin{aligned}
\partial_K \mathcal{J} &= K \int_\rho^1 \frac{s}{(1+2KG_1(s))^2}\, ds - 2K^2 \int_\rho^1 \frac{sG_1(s)}{(1+2KG_1(s))^3}\, ds \\
&= K \int_\rho^1 \frac{s}{(1+2KG_1(s))^3}\, ds,
\end{aligned}$$

which proves (10).

Set $a = 4\sqrt{2}/(3\pi)$. By (9), the change of variable $t = 1 - s$, and $s \le 1$,

$$\begin{aligned}
\int_\rho^1 \frac{s\, ds}{(1+2KG_1(s))^3} &\le \int_0^\infty \frac{dt}{(1+aKt^{3/2})^3} \\
&= \frac{8\pi}{27\sqrt{3}}(aK)^{-2/3}.
\end{aligned} \tag{12}$$

The last equality is the beta integral

$$\frac{2}{3}(aK)^{-2/3} B\left(\frac{2}{3}, \frac{7}{3}\right) = \frac{8\pi}{27\sqrt{3}}(aK)^{-2/3}.$$

At $K = 12$, the cube of the last member of (12) is

$$\frac{\pi^5}{59049\sqrt{3}}.$$

Using (7),

$$\frac{\pi^5}{59049\sqrt{3}} < \frac{5153632}{1654060905} < \left(\frac{88}{567}\right)^3,$$

where the second strict inequality is the exact identity

$$\left(\frac{88}{567}\right)^3 - \frac{5153632}{1654060905} = \frac{27812576}{44659644435} > 0.$$

The right side of (12) decreases with K, proving (11). □

Proposition 3 (Monotonicity of the scalar gap)**.** *For every $\rho_c \le \rho \le 39/50$ and every $K \ge k_{11}(\rho)$,*

$$\partial_K \mathcal{S}(\rho, K) > 0. \tag{13}$$

Consequently,

$$\mathcal{S}(\rho, K) \ge \mathcal{S}(\rho, k_{11}(\rho)). \tag{14}$$

Proof. Write

$$\theta := \arccos \rho, \qquad y := K G_1(\rho).$$

The derivative of the final line of (4) is

$$\frac{2\pi\rho}{\theta}\left(\frac{y(1+y)}{2(1+2y)^2} + \frac{3}{32}\right) \ge \frac{3\rho}{8}, \tag{15}$$

because $\theta \le \pi/2$. Also

$$-\frac{\pi\rho}{4(1-\rho)} > -\frac{11\rho}{14(1-\rho)}.$$

Every $K \ge k_{11}(\rho)$ in the stated ratio interval is greater than 12; this also follows from the stronger estimate established in Lemma 4 below. Lemma 2 therefore gives

$$\begin{aligned}
\partial_K \mathcal{S} &> K\left(\frac{1+2\rho^2}{6} - \frac{88}{567}\right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\
&= K\left(\frac{13}{1134} + \frac{\rho^2}{3}\right) + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)} \\
&> \frac{26}{189} + 4\rho^2 + \frac{3\rho}{8} - \frac{11\rho}{14(1-\rho)}.
\end{aligned} \tag{16}$$

It remains to check the sign of the last expression. Multiplication by $1-\rho$ produces

$$p(\rho) := (1-\rho)\left(\frac{26}{189} + 4\rho^2 + \frac{3\rho}{8}\right) - \frac{11\rho}{14}. \tag{17}$$

The bound $\pi < 22/7$ gives $\rho_c > 7/51$. On the larger rational interval $[7/51, 39/50]$, set

$$u := \frac{\rho - 7/51}{39/50 - 7/51} \in [0, 1].$$

In the degree-three Bernstein basis $B_{j,3}(u) = \binom{3}{j} u^j (1-u)^{3-j}$, exact expansion gives

$$p(\rho) = \sum_{j=0}^{3} b_j B_{j,3}(u), \tag{18}$$

with

$$\begin{array}{c|c}
j & b_j \\
\hline
0 & \dfrac{335005}{2785671} \\
1 & \dfrac{32947387}{196635600} \\
2 & \dfrac{281783183}{578340000} \\
3 & \dfrac{231517}{13500000}
\end{array} \tag{19}$$

All four coefficients are positive. Since the Bernstein basis functions are nonnegative and sum to one, $p(\rho) > 0$. Equation (16) proves (13), and (14) follows by integration in K. $\square$

3 An exact rational upper bound for the loss

Lemma 4 (Uniform loss bound)**.** *For every $\rho_c \le \rho \le 39/50$ and every $K \ge k_{11}(\rho)$,*

$$K > \frac{241}{20} \tag{20}$$

and

$$\int_\rho^1 \frac{x}{(1+2KG_1(x))^2}\,dx < \frac{17}{100}. \tag{21}$$

Proof. Since $\rho \ge \rho_c$,

$$\frac{\pi}{1-\rho} \ge \frac{\pi}{1-\rho_c} = \pi + \frac{1}{2} > \frac{91}{25}.$$

Consequently

$$K^2 \ge k_{11}(\rho)^2 > 132 + \left(\frac{91}{25}\right)^2 = \left(\frac{241}{20}\right)^2 + \frac{471}{10000},$$

which proves (20). The rational bounds in (7) now give

$$\frac{4\sqrt{2}K}{3\pi} > \frac{4(140/99)(241/20)}{3(22/7)} = \frac{23618}{3267} = \frac{36}{5} + \frac{478}{16335}. \tag{22}$$

Put $y = 1 - x$. Equations (9) and (22) yield

$$\begin{aligned}\int_\rho^1 \frac{x\,dx}{(1+2KG_1(x))^2} &< \int_0^{1-\rho} \frac{1-y}{(1+(36/5)y^{3/2})^2}\,dy \\ &\le \int_0^1 \frac{1-y}{(1+(36/5)y^{3/2})^2}\,dy.\end{aligned} \tag{23}$$

We bound the last integral by exact rational arithmetic. Set

$$c := \frac{36}{5}, \qquad \phi(z) := \frac{1}{(1+z)^2},$$

and use the rational grid

$$(t_i)_{i=0}^{10} = \left(0, \frac{1}{4}, \frac{5}{16}, \frac{3}{8}, \frac{7}{16}, \frac{1}{2}, \frac{9}{16}, \frac{5}{8}, \frac{3}{4}, \frac{7}{8}, 1\right). \tag{24}$$

For $z_i = ct_i^3$, let

$$\beta_i := \frac{\phi(z_{i+1}) - \phi(z_i)}{z_{i+1} - z_i}, \qquad \alpha_i := \phi(z_i) - \beta_i z_i.$$

Because $\phi''(z) = 6(1+z)^{-4} > 0$, its secant satisfies

$$\phi(z) \le \alpha_i + \beta_i z \qquad (z_i \le z \le z_{i+1}).$$

After $y = t^2$, the integral in (23) is

$$\int_0^1 2t(1-t^2)\phi(ct^3)\,dt.$$

Thus it is at most

$$Q := \sum_{i=0}^{9} \big(F_i(t_{i+1}) - F_i(t_i)\big), \tag{25}$$

where the exact polynomial antiderivative is

$$F_i(t) := \alpha_i \left(t^2 - \frac{t^4}{2} \right) + 2c\beta_i \left(\frac{t^5}{5} - \frac{t^7}{7} \right). \tag{26}$$

Every number in (25) is rational. Substitution of the grid (24) gives

$$Q = \frac{1305767238061446154064110434835938779253983373178558326887}{7706446260271137290663804190970969288616921855152043745280}, \tag{27}$$

and direct subtraction gives

$$\frac{17}{100} - Q = \frac{2164313092323592674368138814562999905446671098644554905 3}{3853223130135568645331902095485484644308460927576021872640 0} > 0. \tag{28}$$

Equations (23)–(28) prove (21). $\square$

4 The base curve

Fix ρ and now set

$$t := \sqrt{1-\rho}, \qquad k := k_{11}(\rho) = \sqrt{132 + \frac{\pi^2}{t^4}}, \qquad \rho = 1 - t^2. \tag{29}$$

The middle-ratio range lies strictly inside the rational interval

$$\frac{7}{15} < t < \frac{14}{15}. \tag{30}$$

Indeed,

$$t^2 \geq \frac{11}{50} > \left(\frac{7}{15} \right)^2,$$

while $\rho \geq \rho_c > 7/51$ gives

$$t^2 = 1 - \rho < \frac{44}{51} < \left(\frac{14}{15} \right)^2.$$

Lemma 5 (Analytic endpoint payment)**.** *For every $\rho_c \leq \rho \leq 39/50$, let t and k be as in (29), and let*

$$C(\rho) := \frac{100\rho^2 - 1}{600} \tag{31}$$

and

$$E(t) := \frac{11}{14t^2} - \frac{14}{15(1+2t)} - \frac{3}{8t}. \tag{32}$$

Then $C(\rho) > 0$, $E(t) > 0$, and

$$\mathcal{S}(\rho, k) > C(\rho)k^2 - \rho E(t)k + \frac{1}{2}. \tag{33}$$

Proof. First, $\rho \geq \rho_c > 7/51 > 1/10$, so $C(\rho) > 0$. Lemma 4 and

$$\frac{1 + 2\rho^2}{12} - \frac{17}{200} = \frac{100\rho^2 - 1}{600}$$

show that the first two terms in (4) are strictly greater than $C(\rho)k^2$.

Set $\theta = \arccos \rho$ and $Y = kG_1(\rho)$. For $0 \le t \le 1$,

$$\theta = \arccos(1 - t^2) = 2 \arcsin \frac{t}{\sqrt{2}} \le \frac{\pi t}{2}. \tag{34}$$

To see the last inequality, use convexity of arcsin on $[0, 1/\sqrt{2}]$: its graph lies below the chord joining $(0, 0)$ to $(1/\sqrt{2}, \pi/4)$. Furthermore, (9) and $k > \pi/t^2$ give

$$Y > \frac{2\sqrt{2}}{3} t > \frac{14}{15} t. \tag{35}$$

Since $Y \mapsto Y/(1 + 2Y)$ is increasing, (34)–(35) imply

$$\begin{aligned} \frac{2\pi\rho k}{\theta} & \left(\frac{Y}{4(1 + 2Y)} + \frac{3}{32} \right) \\ & > \rho k \left(\frac{14}{15(1 + 2t)} + \frac{3}{8t} \right). \end{aligned} \tag{36}$$

Also

$$-\frac{\pi\rho k}{4t^2} > -\frac{11\rho k}{14t^2}.$$

Combining these inequalities with (4) proves (33).

Finally,

$$840t^2(1 + 2t)E(t) = 660 + 1005t - 1414t^2. \tag{37}$$

The quadratic on the right is concave, so its minimum on $[7/15, 14/15]$ occurs at an endpoint. Its endpoint values are

$$\frac{184739}{225}, \qquad \frac{82406}{225},$$

respectively. Both are positive, proving $E(t) > 0$. $\square$

Proposition 6 (Rational positivity on the base curve)**.** *For every $\rho_c \le \rho \le 39/50$,*

$$\mathcal{S}(\rho, k_{11}(\rho)) > 0. \tag{38}$$

Proof. The elementary inequality

$$\sqrt{x^2 + 132} < x + \frac{66}{x} \qquad (x > 0)$$

and (7) give

$$k^2 > 132 + \frac{(157/50)^2}{t^4}, \tag{39}$$

$$k < \frac{22}{7t^2} + \frac{3300t^2}{157}. \tag{40}$$

Since $C(\rho) > 0$ and $\rho E(t) > 0$, Lemma 5 yields

$$\mathcal{S}(\rho, k) > R(t), \tag{41}$$

where

$$R(t) := C(1-t^2)\left(132 + \frac{(157/50)^2}{t^4}\right) - (1-t^2)E(t)\left(\frac{22}{7t^2} + \frac{3300t^2}{157}\right) + \frac{1}{2}. \tag{42}$$

It remains only exact polynomial algebra. Put

$$H(t) := t^4(1+2t)R(t).$$

Using (37), its factorized form is

$$\begin{aligned} H(t) = C(1-t^2)(1+2t)&\left(132t^4 + \left(\frac{157}{50}\right)^2\right) \\ &- \frac{(1-t^2)(660+1005t-1414t^2)}{840}\left(\frac{22}{7} + \frac{3300}{157}t^4\right) \\ &+ \frac{1}{2}t^4(1+2t). \end{aligned} \tag{43}$$

For an independently checkable expansion,

$$\begin{aligned} H(t) = &-\frac{20642567}{24500000} - \frac{6205067}{12250000}t + \frac{547983}{122500}t^2 - \frac{1033727}{367500}t^3 \\ &+ \frac{34911551}{16485000}t^4 + \frac{187093801}{8242500}t^5 + \frac{8679}{1099}t^6 \\ &- \frac{138149}{2198}t^7 - \frac{2101}{157}t^8 + 44t^9. \end{aligned} \tag{44}$$

Set

$$u := \frac{t - 7/15}{7/15} \in [0,1].$$

In the degree-nine Bernstein basis,

$$H(t) = \sum_{j=0}^{9} d_j B_{j,9}(u), \qquad B_{j,9}(u) = \binom{9}{j} u^j (1-u)^{9-j}, \tag{45}$$

where exact conversion of (44) gives

j	d_j
0	$\frac{65408801807437}{9463832437500000}$
1	$\frac{299549381760793}{1135659892500000}$
2	$\frac{1590469980572077 3}{28391497312500000}$
3	$\frac{2765605256590021}{3154610812500000}$
4	$\frac{33203943051258761}{28391497312500000}$
5	$\frac{38636221171331747}{28391497312500000}$
6	$\frac{2538003206646073}{1892766487500000}$
7	$\frac{28758289199580211}{28391497312500000}$
8	$\frac{12920786461478803}{28391497312500000}$
9	$\frac{342873285597167 9}{9463832437500000}$

(46)

Every d_j is positive. The Bernstein basis functions are nonnegative and sum to one; hence $H(t) > 0$ throughout $[7/15, 14/15]$. Since $t^4(1+2t) > 0$, this proves $R(t) > 0$. Equation (41) proves (38). $\square$

Proof of Theorem 1. Proposition 3 reduces every point of the upward-closed middle-ratio region to the base curve $K = k_{11}(\rho)$. Proposition 6 proves strict positivity on that curve. Therefore $\mathcal{S}(\rho, K) > 0$ throughout $\mathcal{R}_c$, with no upper restriction on K. $\square$

Remark 7 (Nature of the certificate). The rational secants and Bernstein expansions above are finite exact algebra, not interval arithmetic: convexity proves the functional majorant on each secant interval, and positivity follows from explicitly displayed rational coefficients. No floating-point evaluation, sampled minimum, or numerical root isolation is used as a premise.